

Mind Garden

Project Plan

AI-Powered Student Wellness Journal

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1. Problem Statement

academic journey. An enormous workload, strict deadlines, exams, rivalry, professional doubt, and social pressures usually cause stress, anxiety, mood swings, and lack of motivation. There is a lack of professional psychological support services, which is why many students do not actively seek them because of unavailability, high cost, time-related issues, or fears of privacy and stigma.

Despite the numerous existing wellness and mental-health mobile apps, the majority of them target general audiences and include general advice instead of individual assistance. Such platforms are not normally associated with active emotional tracking, dynamic feedback, and substantive interaction. Besides, they do not pay much attention to the daily academic setting of the university students and offer organized instruments of reflective emotional expression like directed journaling and mood monitoring.

The other weakness is that the existing solutions may tend to decouple emotional expression and practical assistance. The students can express their sentiments in personal notes or social media and do not get organized interpretation and constructive suggestions. As a result, the initially connected emotional burnout or long-term stress or deteriorating well-being might remain unnoticed until they have a negative impact on academic performance and personal life.

Thus, there is a necessity of a unified, web-based solution that would allow the students to share their emotions safely, automatically analyze their emotional condition through the methods of Artificial Intelligence, and provide individually customized wellness suggestions. The system must also monitor emotional trends with time and provide meaningful reports, which educate users to conceptualize the behavioral and emotional change.

To fill this gap, the proposed project, Mind Garden Website, will offer a safe and accessible site that will integrate emotional journaling, AI-powered emotion analysis, personal wellness

activities, and progress monitoring. The system also aims to promote the psychological well being of the students by encouraging them to engage in frequent emotional reflection and providing them with adaptive advice that would enable them have a better balance in life between academic and personal life.

2. Outcome

At the project completion, the site will:

- Enabling users to create and log-in safely.
- Allow users to make daily entries, which are emotional and written in both text and emojis.
- Identify emotions to include stress, anxiety, sad, calm, positive and motivated.
- Offer customized wellness instructions (breathing exercises, mindfulness, writing prompts, podcasts).
- Log moods of the users over time and create weekly/monthly reports.
- Enable user to change profile and preferences.

3. Requirements

Functional Requirements

- User Authentication (Register/Login)
- Mood input (text + emojis)
- AI emotion analysis
- Personalized recommendations
- Progress reports

Non-Functional Requirements

- Response time ≤ 3 seconds
- Secure communication (HTTPS)
- User-friendly, English-only interface
- Compatible with major browsers
- Scalable backend architecture

Technical Specifications

Spec	Description
Platform	Web Application (English)
Frontend	HTML, CSS, JS
Backend	Python FastAPI/Flask
AI Model	English NLP for emotion classification
Database	MySQL/SQLite
Security	JWT, encrypted storage
Testable Metrics	Response time, recommendation accuracy, report generation

4. Relevance and Impact

The students at the universities are one of the most susceptible groups to mental and emotional stress. Academic stress, time limits, test, financial stability, and social demands have a great impact on their mental health. In accordance with the recent international educational and psychological research, a substantial proportion of university students report to have anxiety, burnout, and mood instability throughout their college years.

The existing mental wellness applications in the web are currently limited in a number of ways:

- They are not student friendly.

- They have not been culturally adjusted.
- They are not personalized and are generic.
- And they lack continuous emotional monitoring.

The Mind Garden Website addresses these problems with the goal of helping students solve them by offering an emotional support platform intending to be student-focused and which combines journaling, AI-based emotion analysis, and personalized wellness recommendations.

It is expected to have the following effect:

- Getting students more familiar with their emotional conditions.
- Promoting self-reflection and expression of emotions every day.
- Lessening stress with facilitated wellness programs.
- Giving early warning on emotional burnout.
- Enhancing the academic productivity and psychological stability of students.

The university environment has hundreds of thousands of university students who may be benefited by the system. The platform can be extended to serve several universities and educational institutions in the long run.

The importance of this solution is high since the awareness of mental health is growing globally, and the accessibility of digital support tools can have a positive effect on the academic levels of students as well as their personal growth and quality of life.

5. Advisor

Name of faculty member supervising the project.

6. Team and Roles

Sr. #	ID	Name	Roles
1	2221005417	Lamya Faisal Alahmari	Project Manager (team coordination, submission of deliverables)
2	2221001705	Fajr Fahad Alshammri	Requirement Analyst (requirements gathering, SRS documentation)
3	2221003976	May Mazah Aldafeeri	System Architect (system architecture design, AI integration planning)
4	2221001396	Dana Mohammed Al-Anzi	System Designer (design, database structure design)
5	2221002540	Reham Mused	Test Engineer
6	2221005376	Atheer Abdullah Alfaifi	Development Engineer (backend development – API & database implementation)
7	2221005503	Aryam Ayed Al-Rashidi	Frontend Developer (interface, user interaction, responsiveness)
8	2221003991	Bashayer Melfi Alrsheedi	AI & Integration Developer (emotion analysis model, recommendation engine integration)

7. Timetable

Task ID	Task Name / Description	Owner	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1.0	Requirements Phase	Fajr (Requirement Analyst)	X	X													
1.1	Literature Review	Fajr	X	X													
1.2	Identify System Requirements	Fajr + Lamya		X	X												
1.3	SRS Documentation	Lamya			X	X											
2.0	System Design Phase	May (System Architect)			X	X	X										
2.1	Database Design	Dana			X	X											
2.2	Prototype	Dana + Aryam				X	X	X									
2.3	System Architecture	May			X	X	X										
3.0	Development Phase	Atheer (Backend Dev)					X	X	X	X	X						
3.1	Backend API Development	Atheer					X	X	X	X							
3.2	Frontend Development	Aryam						X	X	X	X						
3.3	AI Emotion Analysis Model	Bashayer					X	X	X	X	X						

4.0	Integration Phase	May + Bashayer							X	X	X						
4.1	Connect Frontend & Backend	Atheer + Aryam							X	X							
4.2	Integrate AI Model	Bashayer								X	X						
5.0	Testing Phase	Reham (Test Engineer)									X	X	X				
5.1	Unit Testing	Reham									X	X					
5.2	System Testing	Reham										X	X				
5.3	Bug Fixing	Atheer + Aryam											X	X			
6.0	Deployment Phase	Lamya												X	X		
6.1	Server Deployment	Lamya + Atheer												X			
6.2	Final System Verification	Reham													X		
7.0	Documentation & Presentation	All Team													X	X	X
7.1	Final Report Writing	All Team													X	X	
7.2	Presentation Slides	All Team														X	X
7.3	Final Demonstration Preparation	All Team															X